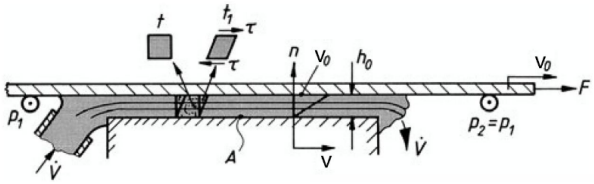
I. Viscous flow

Symbology

Symb	Name	Units	Symb	Name	Units
ν	Kinematic viscosity	m^2/s	η	Dynamic viscosity	Pa·s
τ	Shear stress	Pa	dn	Normal distance	m
du/dy	Rate of strain	[-]	e_{Diss}	Spec. diss. power	W/m^3

1. Flows with friction



$$F = \eta \frac{v_0 \cdot A}{h_0} \implies \tau = \frac{F}{A} = \eta \frac{v_0}{h_0} = \eta \frac{dv}{dn}$$

For pipes:

$$F_\tau = \tau A \implies M_\tau = F_\tau y = \tau \cdot l \cdot 2\pi r_0^2$$

1.1 Friction law

$$\tau = -\eta \frac{dv}{dn} \quad ; \quad \dot{e}_{\text{Diss}} = \eta \left(\frac{dv}{dn}\right)^2$$

2. Viscosity

If $\eta = 0$, the fluid is called inviscid.

No slip condition: velocity of a particle closest to the wall has velocity equal to the wall velocity.

2.1 Reynolds Number

$$Re = \frac{\rho v L}{\eta} = \frac{vL}{\nu} \quad ; \quad \nu = \frac{\eta}{\rho}$$

2.2 Newtonian fluids

- Rate of deformation $\mathrm{d}u/\mathrm{d}y$ (velocity gradient) is linearly proportional to the shear stress τ
- The constant of proportionality is the viscosity (slope)
- $\eta = \dfrac{\tau}{\mathrm{d}u/\mathrm{d}y} = \text{constant}$

2.3 Non-Newtonian fluids

Non-linear relation between shear stress and deformation rate due to non-constant viscosity.

$$\tau = \eta \left(\frac{\partial u}{\partial y}\right) \cdot \frac{\partial u}{\partial y}$$

2.3.1 Dilatant (shear thickening) fluids

Viscosity increases with increasing strain rate.

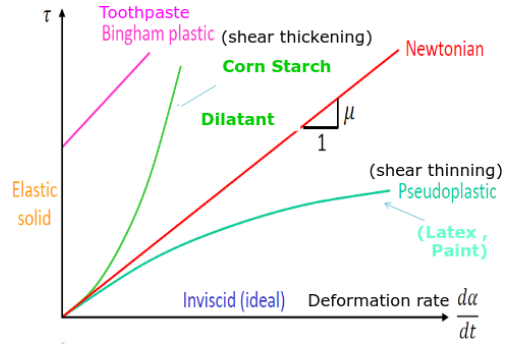
2.3.2 Plastic or pseudo-plastic (shear thinning) fluids

Viscosity decreases with increasing strain rate.

2.3.3 Bingham medium

Flow occurs only after a yield stress τ_0 is reached.

2.4 Graphical representation



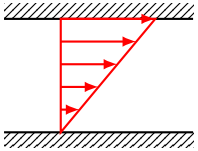
3. Flow profiles

3.1 Couette flow profiles (constant velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{v_p}{s}$$

$$e_{\text{Diss}} = \eta \left(\frac{\partial u(y)}{\partial y}\right)^2 = \eta \left(\frac{v_p}{s}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}(y)\, dV$$



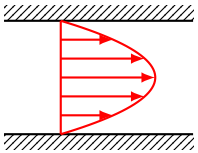
3.2 Poiseuille flow profiles (parabolic velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{4v_{\text{max}}}{s}\left(1 - \frac{2y}{s}\right)$$

$$v(r) = ar^2 + br + c$$

$$e_{\text{Diss}} = \eta \frac{16 v_{\text{max}}^2}{s^2}\left(1 - \frac{2y}{s}\right)^2$$

$$P_{\text{Diss}} = \frac{16\eta A v_{\text{max}}^2}{3s}$$



4. Laminar pipe flow

4.1 $p_1 > p_2$

$$\frac{\partial p}{\partial z} = \frac{p_2 - p_1}{L}$$

4.2 Constant linear flow profiles

$$\dot{V} = v_m A = v_m R^2 \pi$$

4.3 Parabolic flow profiles

$$\dot{V} = \int v(r)\, dA = \frac{v_{max} \cdot R^2 \pi}{2}$$

$$v(r) = v_{max}\left(1 - \frac{r^2}{R^2}\right)$$

$$\frac{dv}{dr} = -2v_{max}\frac{r}{R^2}$$

$$P_{\text{Diss}} = \int_0^L \int_0^{2\pi} \int_0^R \eta \left(-2v_{max}\frac{r^2}{R^2}\right) \cdot r\, dr\, d\varphi\, dx$$

$$P_{\text{Diss}} = 2\pi L \int_0^R \eta \left(-2v_{max}\frac{r^2}{R^2}\right) \cdot r\, dr = 2\pi L \eta v_{max}^2 = 8\pi L \eta v_m^2$$

5. Flows in gaps and bearings

5.1 Assumptions in gaps and bearings

- Newtonian fluid with constant viscosity η
- Incompressible fluid
- Very low Reynolds number $Re < 2000$
- Laminar flow
- Balance of pressure and viscous forces
- No slip condition at the walls
- No acceleration (negligible)
- Pressure is uniform across the thickness $\partial p/\partial y \approx 0$

5.2 Equilibrium of forces

$$\sum F_x = 0 \implies pA_p - (p + dp)A_p + \tau A_\tau + (\tau + d\tau)A_\tau = 0$$

$$A_p = dy \cdot b \quad ; \quad A_\tau = dx \cdot b \quad ; \quad \tau = \eta \frac{dv}{dy} \to \frac{d\tau}{dy}$$

$$\boxed{\eta \frac{d^2 v(x,y)}{dy^2} = \frac{dp(x)}{dx} = p'}$$

5.3 Basic velocity equation

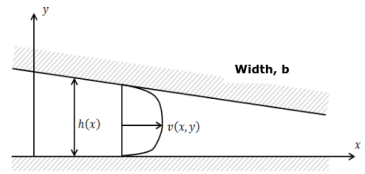
$$v(x,y) = \frac{1}{\eta} \cdot p' \cdot \frac{y^2}{2} + c(x)y + d(x)$$

5.3.1 Gap flow (walls not moving)

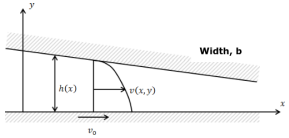
$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx}$$

$$d(x) = 0$$



5.3.2 Plain bearing / Creeping gap flow (one wall moving)



Assumption for $Re \ll 1$: $\eta \dfrac{d^2 v}{dy^2} = \dfrac{dp}{dx} = p'$

$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right] + \frac{v_0}{h(x)}\left[h(x) - y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx} - \frac{v_0}{h(x)}$$

$$d(x) = v_0$$

For the simplest, parabolic velocity profile case:

$$h(x) = h_0 \quad ; \quad v_0 = 0 \quad ; \quad p' = \frac{dp(x)}{dx} = \frac{\Delta p}{\Delta L} = -\frac{12\eta v_m}{h_0^2}$$

$$v(y) = \frac{p'}{2\eta}\left(y^2 - h_0 y\right) \Longrightarrow \frac{\partial v}{\partial y} = 0 = \frac{p'}{2\eta}\left(2y - h_0\right)$$

$$y = \frac{h_0}{2} \Longrightarrow v_{max} = v\left(y = \frac{h_0}{2}\right) = -\frac{h_0^2 p'}{8\eta} = \frac{3}{2} v_m$$

$$v_m = -\frac{\Delta p \cdot h_0^2}{12\eta L}$$

$$\dot{V} = b\int_0^{h_0} v(y)\,dy = -b\frac{p' h_0^3}{12\eta} = \frac{h_0^2}{6\eta} b\Delta p$$

General sliding bearings:

$$\dot{V} = b\int_0^{h(x)} \frac{p'}{2}\eta\left(y^2 - h(x)y\right) + \frac{v_0}{h(x)}\left(h(x) - y\right)\,dy$$

$$\dot{V} = -b\frac{p' h_{(x)}^3}{12\eta} + b\frac{v_0 h(x)}{2} = ¢$$

$$p' = 6\eta v_0\left(\frac{1}{h(x)^2} - \frac{\dot{V}}{b h(x)^3}\right)$$

5.3.3 General form of the volumetric flow rate

$$v(x,y) = \frac{1}{\eta} p' \frac{y^2}{2} + c(x) + d(x)$$

$$\dot{V} = b\int_0^{h(x)} v(x,y)\,dy = b\left[-\frac{h^3(x)}{6\eta}p' + c(x)\frac{h^2(x)}{2} + d(x)h(x)\right] = ¢$$

$$p' = \frac{6\eta}{h(x)^3}\left(\frac{\dot{V}}{b} - c(x)\frac{h(x)^2}{2} - d(x)h(x)\right)$$

5.3.4 Possible semplifications

If h $\ll$ D:

$$A = \int_A r\,dr\,d\varphi \approx \pi D h$$

5.4 Specific dissipation power

$$P = \tau A v = \eta \frac{\partial v}{\partial n} A v \quad ; \quad dP = dF dv$$

$$dF_\tau = d\tau dA = \eta \frac{dv}{dy} dx dz$$

$$e_{\text{Diss}} = \frac{dP_\tau}{dV} = \frac{\eta \frac{dv}{dy} dx dz dv}{dx dy dz} = \eta\left(\frac{dv}{dy}\right)^2 = \eta\left(\frac{\partial v}{\partial y}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}\,dV = \iiint_V e_{\text{Diss}}\,dx dy dz = M \cdot \omega$$

5.4.1 Pressure loss according to Bernoulli

$$p_1 - p_2 = \rho \frac{P_{\text{Diss}}}{\dot{m}} = \frac{P_{\text{Diss}}}{\dot{V}} = \frac{8\pi L \eta v_m}{R^2} \; (= \rho g h)$$

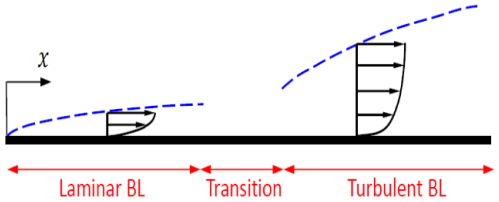
II. Boundary layers

Symb	Name	Units	Symb	Name	Units
v_∞	free-stream velocity	m/s	$v(y)$	local velocity in boundary layer	m/s
δ	boundary layer thickness	m	δ^*	BL displacement thickness	m
θ	momentum thickness	m	τ_w	wall shear stress	Pa
ν	kinematic viscosity	m²/s	η	dynamic viscosity	Pa·s
Re_x	Reynolds num. at position x	-	$\dot{V}_{BL}$	boundary-layer volume flow rate	m²/s

Boundary layers develop due to wall shear stress τ and fluid viscosity ν.

6. Laminar vs. Turbulent boundary layers

6.1 Critical Reynolds number

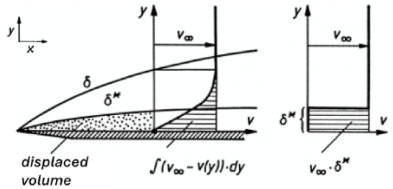


Reynolds number for the BL on a flat plate:

$$Re_x = \frac{v_\infty x}{\nu} = \frac{v_\infty \cdot l_{char}}{\nu}$$

Re **increases** with the distance from the plate entry.
Transition to turbulent: $Re_x \geq 5 \times 10^5$

6.2 Flat plate boundary layer



$$\delta(x) \approx \sqrt{x} \quad ; \quad \tau(x) \approx 1/\delta(x) = 1/\sqrt{x}$$

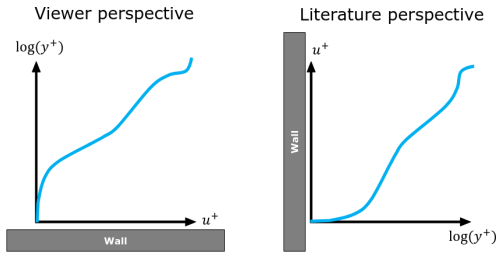
$$\dot{V}_{BL} = v_\infty \cdot \delta^*(x)$$

with 2 plates: $\dot{V}_{BL} = A^* \cdot v = \left(\dfrac{d - 2\delta^*}{2}\right)\pi \cdot v$

6.3 Turbolent BL

6.3.1 Non-dimensional forms

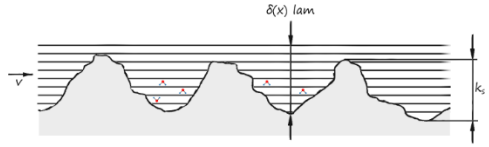
Non-dimensional plot of wall normal turbulent velocity profile:



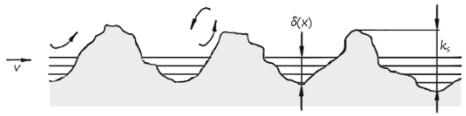
- Friction velocity: $u_\tau = \sqrt{\dfrac{\tau_{\text{wall}}}{\rho}}$
- Dimensionless wall parallel velocity: $u^+ = \dfrac{u}{u_\tau}$
- Dimensionless wall distance: $y^+ = \dfrac{y u_\tau}{\nu}$

6.4 Rough and smooth walls and the effect on friction

6.4.1 No-slip condition



Hydraulically Smooth Surface



Rough Surface

- Hydraulically smooth surface: $\delta(x) > k_s$
- Rough surface: $\delta(x) < k_s$. Leads to vortex.

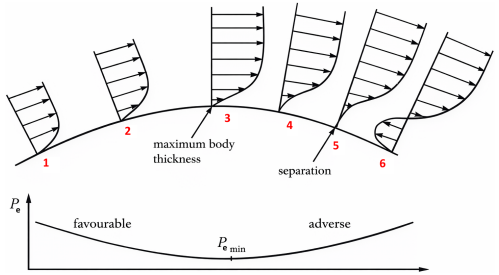
7. Flow separation

7.1 Flow separation conditions

1. Friction
2. Pressure increase

At sharp edges, the flow always separate.

7.2 Pressure gradient



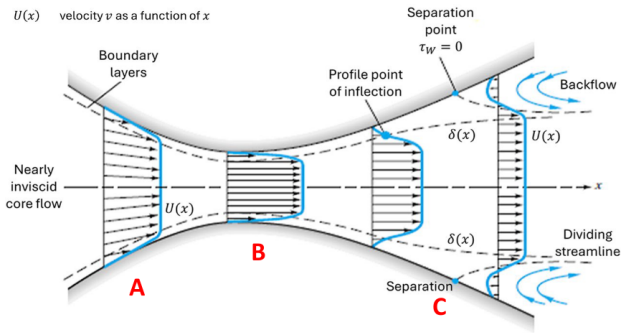
7.2.1 Comparison of boundary layer profile shape

Comparing BL as a function of the pressure gradient:

$$\frac{dP}{dx} \sim -\frac{dU(x)}{dx} \quad ; \quad \tau_{\text{wall}} = m\frac{dv}{dy_{y=0}}$$

1. favorable pressure gradient: $\frac{\partial p}{\partial x} \ll 0$
2. favorable pressure gradient: $\frac{\partial p}{\partial x} < 0$
3. zero pressure gradient: $\frac{\partial p}{\partial x} = 0$
4. mild adverse pressure gradient: $\frac{\partial p}{\partial x} > 0$
5. critical adverse pressure gradient: $\frac{\partial p}{\partial x} > 0, \tau_{\text{wall}} = 0$
6. large adverse pressure gradient: $\frac{\partial p}{\partial x} \gg 0$

7.3 Flow separation gradients



7.3.1 Nozzle (Favorable Pressure Gradient) (A)

$$\frac{dp}{dx} < 0 \quad ; \quad \frac{dU}{dx} > 0$$

- Decreasing pressure and decreasing area
- Pressure decreases in the flow direction
- Pressure force is in the flow direction
- No flow separation

7.3.2 Throat (B)

$$\frac{dp}{dx} = 0 \quad ; \quad \frac{dU}{dx} = 0$$

- Constant pressure and constant area
- Fluid particles in the BL slow down due to shear stress only
- No flow separation can occur
- Similar to a flat plate

7.3.3 Diffuser (Adverse Pressure Gradient) (C)

$$\frac{dp}{dx} > 0 \quad ; \quad \frac{dU}{dx} < 0$$

- Pressure increases in the flow direction
- Pressure force is in opposite direction of the flow
- **Back flow situation**: Fluid particles close to the wall with low momentum may come to a stop or even move in opposite direction of the main flow

7.4 Influence of Re on blunt bodies

- $Re < 1$: No vortex formation
- $Re \approx 10^5$: Laminar BL, vortex formation in dead water area
- $Re > 10^6$: BL partially turbolent, vortex on the body surface

III. Drag and Lift Forces

Drag force: parallel and opposite to the relative flow direction
Lift force: perpendicular to the relative flow direction

8. Lift force and Drag coefficient

8.1 Drag force

$$\boxed{F_D = C_D q_\infty A_P = C_D \frac{1}{2}\rho v_\infty^2 A_p}$$

where:

C_D : drag coefficient (μ) ; A_p : project area
q_∞ : dynamic contribution

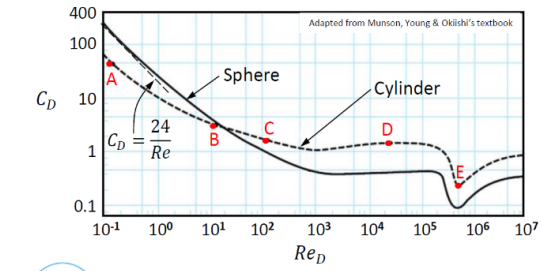
8.1.1 Drag force and power on a body in a flow

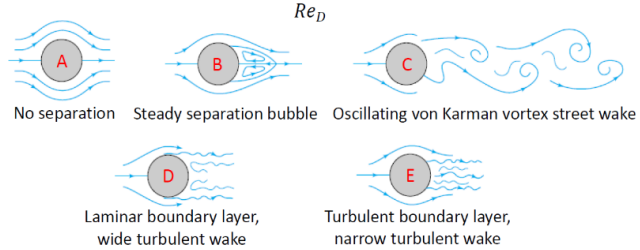
$$\boxed{F_D = F_P + F_\tau \quad ; \quad P_D = F_D v_\infty}$$

Friction drag: friction forces F_τ due to wall shear stress τ_{wall}
Pressure drag: pressure forces F_P due to pressure variation along the surface of the object

8.2 Drag on smooth circular cylinders

8.2.1 Prandlt's experiment





8.2.2 Stokes formula and Stokes law

$$\boxed{F_D \approx F_\tau = 3\pi\eta d v_\infty \Longrightarrow C_D = \frac{24}{Re}}$$

Velocity of a sphere

$$\frac{1}{2}C_D\rho_{\text{air}}Av_\infty = \rho_{\text{sph}}V_{\text{sph}}g \quad ; \quad A = \left(\frac{d}{2}\right)^2\pi \quad ; \quad V = \left(\frac{d}{2}\right)^3\pi$$

$$\boxed{v_\infty = \sqrt{\frac{4}{3}\frac{d\cdot g\cdot\rho_{\text{sph}}}{C_D\cdot\rho_{\text{air}}}}}$$

8.2.3 Free falling sphere

$$F_{res} = F_g - F_n \Longrightarrow F_g = F_n$$

$$mg = C_D A\rho_{\text{fluid}}\frac{v_\infty^2}{2} \Longrightarrow v_\infty = \sqrt{\frac{2mg}{C_D A\rho_{\text{fluid}}}}$$

Velocity profile at the initial phase ($F_{res} \neq 0$)

$$m\frac{dv}{dt} = mg - C_D A\rho_{\text{fluid}}\frac{v^2}{2}$$

$$\int_0^t dt = \int_0^v \frac{m}{mg - C_D A\rho_{\text{fluid}}\frac{v^2}{2}}\,dv$$

$$v(t) = v_\infty \tanh\frac{gt}{v_\infty}$$

Duration of the initial phase

If steady terminal velocity $v(t) = v_\infty \cdot 0.99$:

$$\tanh \frac{gt}{v_\infty} = 0.99 \Longrightarrow \frac{gt}{v_\infty} = \text{arctanh}(0.99)$$

$$t = \frac{v_\infty \, \text{arctanh}(0.99)}{g} = \frac{v_\infty \cdot 2.65}{g}$$